

DANIEL AMEWOALOR

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Department of Biochemistry and Biotechnology

Kwame Nkrumah University of Sci. and Tech.

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EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST) - Kumasi, Ghana January 2021- September 2024
B.Sc. in Biochemistry, Graduated *Summa Cum Laude* (First Class Honors, CGPA: 3.92/4.0)

Dissertation Title: Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis

Advisor: Alexander Kwarteng, Ph.D.

RESEARCH EXPERIENCE

Graduate Intern, Kumasi Centre for Collaborative Research in Tropical Medicine September 2025 – October 2025

Principal Investigators: Dorcas Owusu, Ph.D and Mr. Dennis Albert Kegya

Project Titles: Non-tuberculous *Mycobacteria* in pulmonary disease patients with presumptive tuberculosis from Ghana; artemisinin resistance surveillance study in Sub-Saharan Africa

- Isolated peripheral blood mononuclear cells (PBMCs) from blood samples using Ficoll-based density gradient centrifugation
- Examined and quantified PBMCs using trypan blue stain, the Neubauer chamber, and the bright field microscope
- Cultured and monitored the growth of *Plasmodium falciparum* using RPMI media, hematocrit, and malaria microscopy
- Performed ring stage survival assay to evaluate the in vitro susceptibility of *P. falciparum* to artemisinin following Percoll and sorbitol synchronization experiments

Research Assistant, Human Genetics and Genomics Lab, KNUST

July 2025 – December 2025

Principal Investigator: Lord J.J. Gowans, Ph.D.

Project Title: Deciphering the genetic and genomic architecture of neurodevelopmental disorders in Ghana using whole genome sequencing

- Led participant recruitment and sample collection at the Pediatric Neurology Clinic, Komfo Anokye Teaching Hospital – Kumasi, after obtaining informed consent and processing whole blood into EDTA tubes and buffy coats with full traceability
- Managed laboratory sample inventory for 150 samples by implementing a tracking system to streamline preprocessing and downstream genetic analysis workflows
- Extracted genomic DNA from whole blood samples and buffy coat using silica-based spin columns
- Quantified the concentration of DNA using the Qubit Fluorimeter and assay

Independent Investigator, KNUST

September 2024 – June 2025

Project Title: Integrative cancer genomics approach to design and characterize a multi-epitope mRNA vaccine construct targeting oncogenes *CDC25C* and *AURKA* in hepatocellular carcinoma

- Identified and selected oncogene targets (*CDC25C*, *AURKA*) from the TCGA-LIHC dataset using DESeq2 (Bioconductor, R) for differential expression analysis and TISCH2 for single-cell validation; mapped MHC-I and MHC-II binding and immunogenic epitopes using immunoinformatics tools (IEDB, VaxiJen, AllerTop, ToxinPred)
- Performed structural modeling of the vaccine peptide using AlphaFold2
- Conducted molecular dynamics simulations to assess the stability and flexibility of the vaccine-TLR2 complex using GROMACS and analysed the trajectories with the MDAnalysis Python package
- Executed computational immune simulation using the C-IMMSIM bioinformatic server to predict B cell, T cell, and memory responses

Lead Student Investigator, KNUST

January 2024 – September 2024

Principal Investigator: Alexander Kwarteng, Ph.D.**Project Title:** Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis

- Conceptualized and led the project to identify and validate a novel vaccine target from the filarial parasite, performing comprehensive B cell and T cell epitope mapping using IEDB prediction tools
- Constructed the primary and secondary structure of the multi-epitope subunit vaccine with PSIPRED and modeled the tertiary structure for docking studies
- Performed protein-protein docking with TLR4 using ClusPro and analyzed binding interactions to evaluate vaccine candidate recognition by the innate immune system
- Simulated *in silico* immune responses with C-IMMSIM webserver to quantify antibody titers, cytokine production, and long-term memory formation

Research Intern, Human Genetics and Genomics Lab, KNUST

October- December 2023

Principal Investigator: Lord J.J. Gowans, Ph.D.**Project Title:** A Genome-wide scan for genetic loci associated with Alzheimer's disease among Ghanaians

- Collected and processed saliva samples using Oragene sample collection kits, adhering to protocols for sample preservation and traceability throughout the research process
- Extracted high-quality genomic DNA from samples obtained from patients with Alzheimer's disease using the Oragene DNA protocol, ensuring sample integrity for downstream genetic analyses
- Quantified DNA concentrations using a Qubit Fluorometer and assay, achieving precise measurements important for accurate genomic studies
- Validated DNA integrity through PCR amplification and agarose gel electrophoresis, ensuring sample suitability for downstream SNP array analysis

Research Intern, Borquaye Research Group, KNUST

October -December 2022

Principal Investigator: Lawrence S. Borquaye, Ph.D.**Project Title:** Elucidating antimicrobial and antioxidant activities of selected plant extracts

- Analyzed protein-ligand interactions between the *Pseudomonas aeruginosa* LasR protein and phytochemicals from *Cryptolepis sanguinolenta* using PyRx, PyMol, and Discovery Studio to identify potential quorum-sensing inhibitors
- Cultured bacteria on nutrient agar and standardized the concentration using the McFarland standard
- Performed antimicrobial susceptibility testing against *K. pneumoniae*, *P. aeruginosa*, *E. faecalis*, and *C. albicans*
- Conducted antioxidant assays such as DPPH and H₂O₂ assays to validate the bioactivities of selected plant extracts

CLINICAL EXPERIENCE

Medical Laboratory Intern

November-December 2021

Saint John of God Hospital, Sefwi-Asafo**Supervisor:** Mr. Philemon Agyemang

- Examined sickle cell disease using the sodium metabisulfite reagent to detect sickling in red blood cells
- Performed sickle cell disease screening using sodium metabisulfite to detect red blood cell sickling
- Determined ABO and Rh blood groups via serological typing, ensuring accuracy for transfusion compatibility
- Conducted hemoglobin variant analysis using cellulose acetate electrophoresis for genotype determination
- Maintained meticulous records of all laboratory data and results in compliance with hospital protocols

WORK EXPERIENCE

Teaching and Research Assistant

October 2024-December 2025

Department of Biochemistry and Biotechnology, KNUST

Supervisor: Dr. Lord J.J. Gowans

- Annotated and prioritized genetic variants associated with craniosynostosis and mandibuloacral dysplasia using Ensembl's VEP, Genecard suite, and other bioinformatic tools to support a collaborative genetics research project
- Assisted in the supervision of five undergraduate thesis groups investigating the prevalence of G6PD deficiency at KNUST by guiding students in experimental design, data collection, and analysis
- Facilitated tutorial sessions and assessed term papers for Molecular Genetics (302 students) and Molecular Biology (260 students)
- Taught undergraduate laboratory courses in biochemistry and molecular biology and evaluated laboratory reports

MANUSCRIPTS AND PREPRINTS

Structured Prereview

Ogunniyi T., Natale N.P., Gyabaah E.A, Poku-Adusei E., **Amewoalor D.**, Savai S.M. "Standard precautions perception and practice among health workers in the obstetrics-gynecology department of a referral hospital in Cameroon".

DOI: [10.5281/zenodo.16933811](https://doi.org/10.5281/zenodo.16933811)

Structured Prereview

Ogunniyi T., Otabil J., Mohammed-Nafi'u R., Brobbey S.S, Essandoh A., **Amewoalor D.**, and Poku-Adusei E. "Early Metabolic Alterations in Cerebrospinal Fluid Fatty Acid Profiles Linked to Cognitive Decline and All-Cause Dementia".

DOI: [10.5281/zenodo.17107906](https://doi.org/10.5281/zenodo.17107906)

Manuscript in Preparation

Amewoalor D., Danquah F.Y.E., Manu K.K., Dunyo P., Abakah A., & Kwarteng A. "Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis."

CONFERENCE / PRESENTATIONS

Poster Presenter

Amewoalor D., Baffoe V.D, Mensah G.O, Martyn-Dickens C., Tsri B., Tweneboah J.O, Asamoah C.O, Sabbah D.K, Tawiah E.A., Plange-Rhule G., Busch T., Obiri-Yeboah S., Butali A., Hing A., Donkor P., Gowans L.J.J. Recurrent pathogenic *LMNA* gene variant causes Mandibuloacral Dysplasia in a Ghanaian multiplex family.

KNUST Scientific Conference (TEKCONFAB'25), November 2025. ([Abstract #306](#), [Abstract Book#306](#), [Poster](#)).

Virtual Poster Presenter

Amewoalor D., *In silico* design and characterisation of a novel multi-epitope mRNA vaccine against hepatocellular carcinoma targeting overexpressed CDC25C and AURKA.

13th Annual Cancer Research & Oncology Virtual Event Series by (Labroots), October 2025. ([Poster](#)).

Poster Presenter

Amewoalor D., Danquah F.Y.E., Manu K.K., Dunyo P., & Kwarteng A., Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis.

Ghana Science Association - Kumasi Annual Conference, November 2024.

Poster Presenter

Amewoalor D., Danquah F.Y.E., Manu K.K., Dunyo P., & Kwarteng A., Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis.

Faculty of Physical and Computational Sciences' PosterFest'24, KNUST, October 2024.

Virtual Poster Presenter

Amewoalor D., Danquah F.Y.E., Manu K.K., Dunyo P., & Kwarteng A., Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis.

Molecular Parasitology Meeting XXXV at Woods Hole, USA, September 2024.

RELEVANT SKILLS

Molecular Biology & Biochemistry

DNA/RNA extraction (silica columns, Oragene), DNA quantification (Qubit), PCR amplification, gel electrophoresis, bacterial culture, cell culture (mammalian and *Plasmodium*), PBMC isolation

Genomics & Data Analysis

R (DESeq2, edgeR, tidyverse, ggplot2, TCGAbiolinks, pheatmap), Python, Linux/Bash, genetic variant analysis (Ensembl VEP, VarElect, STRING, SIFT, Polyphen2, AlphaMissense, ClinVar, ClinPred, gnomAD, OMIM, HOPE), multiple sequence alignment (Clustal Omega)

Cancer Research & Immunology

Immunoinformatics (IEDB, VaxiJen, AllerTop), immune simulation (C-IMMSIM), epitope design, cancer genomics, and oncogene targeting (GEPIA, cBioPortal, Tumor Immune Single-cell Hub 2 (TISCH2))

Structural & Computational Biology

AlphaFold2, GalaxyRefine, PyMol, Chimera, PyRx, Discovery Studio, ClusPro, UniProt, *in silico* cloning (SnapGene, GenScript)

Microscopy

Bright field microscopy, malaria microscopy, cell counting (Neubauer chamber, trypan blue)

SCHOLARSHIPS AND AWARDS

Ghana Science Association	
KNUST Bursary Award	2023 -2024
Vice Chancellor’s competitive award for brilliant but needy students	
KNUST Students’ Representative Council Bursary	2022- 2023
A competitive tuition scholarship awarded to brilliant but needy students	
Cocoa Scholarship Award	2017- 2019
Ghana Cocoa Board	
Awarded a full academic scholarship to a public high school in Ghana	

PROFESSIONAL TRAINING WITH CERTIFICATIONS

• Bioinformatics for Biologists: an Introduction to Linux, Bash Scripting, and R Online Course by Wellcome Connecting Science	June 2025
• Biomedical Responsible Conduct of Research by Collaborative Institutional Training Initiative Program	October 2024
• Human Subjects Learners by Collaborative Institutional Training Initiative Program	October 2024
• Professional development training workshop for oncology team by Clinical Oncology Training Centre Workshops on multidisciplinary approaches in oncology to enhance patient outcomes	October 2024
• “R Programming for Data Science ” by Stanford Medicine	July 2024
• Summer Computational Biochemistry Training by GYRP Virtual workshops on computational protein design and analysis	July 2024
• “Cancer Biology Specialization” by Johns Hopkins University	November 2023
• “Introduction to Breast Cancer” by Yale University	November 2023
• “Immunology: Immune Failures and Cancer Immunology” by Imperial College London	November 2023
• “Bioinformatic Methods I & II ” by University of Toronto	October 2023
• “NDG Linux Unhatched” by the Cisco Networking Academy	October 2023

PROFESSIONAL AFFILIATION

Ghana Science Association	November 2024 - Present
Ghana Youth Research Program (GYRP)	January 2023- Present
Genetics Society of West Africa	December 2023- Present

LEADERSHIP, COMMUNITY INVOLVEMENT AND VOLUNTARY WORK

Ghana Biochemistry Students' Association ♦ *Academic Committee member* January- September 2023

- Organized a year-long mentorship scheme to support students with academic challenges
- Assisted in organizing the Prof. K. Nsiah Quiz and Debate Competition in honor of Prof. K.Nsiah

Science Students' Association ♦ *Publicity Team Member* January- September 2023

- Coordinated participation in college-wide events (homecoming, orientation, and recruitment fairs)
- Distributed press releases and promotional materials (flyers, posters) for events and activities

REFERENCES

Dr. Lord J.J. Gowans

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Dr. Antonia Yarney Tetteh

Associate Professor

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